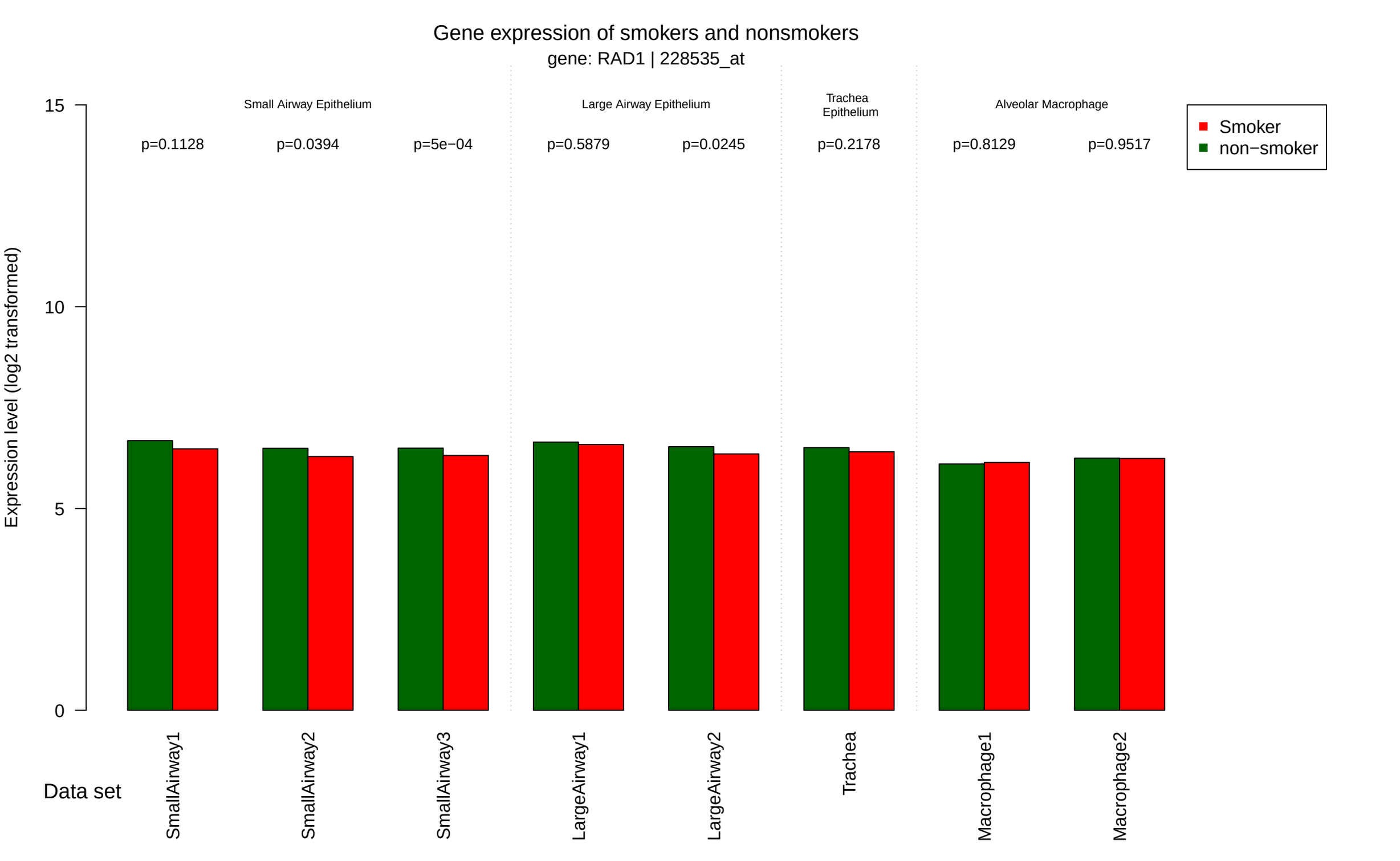
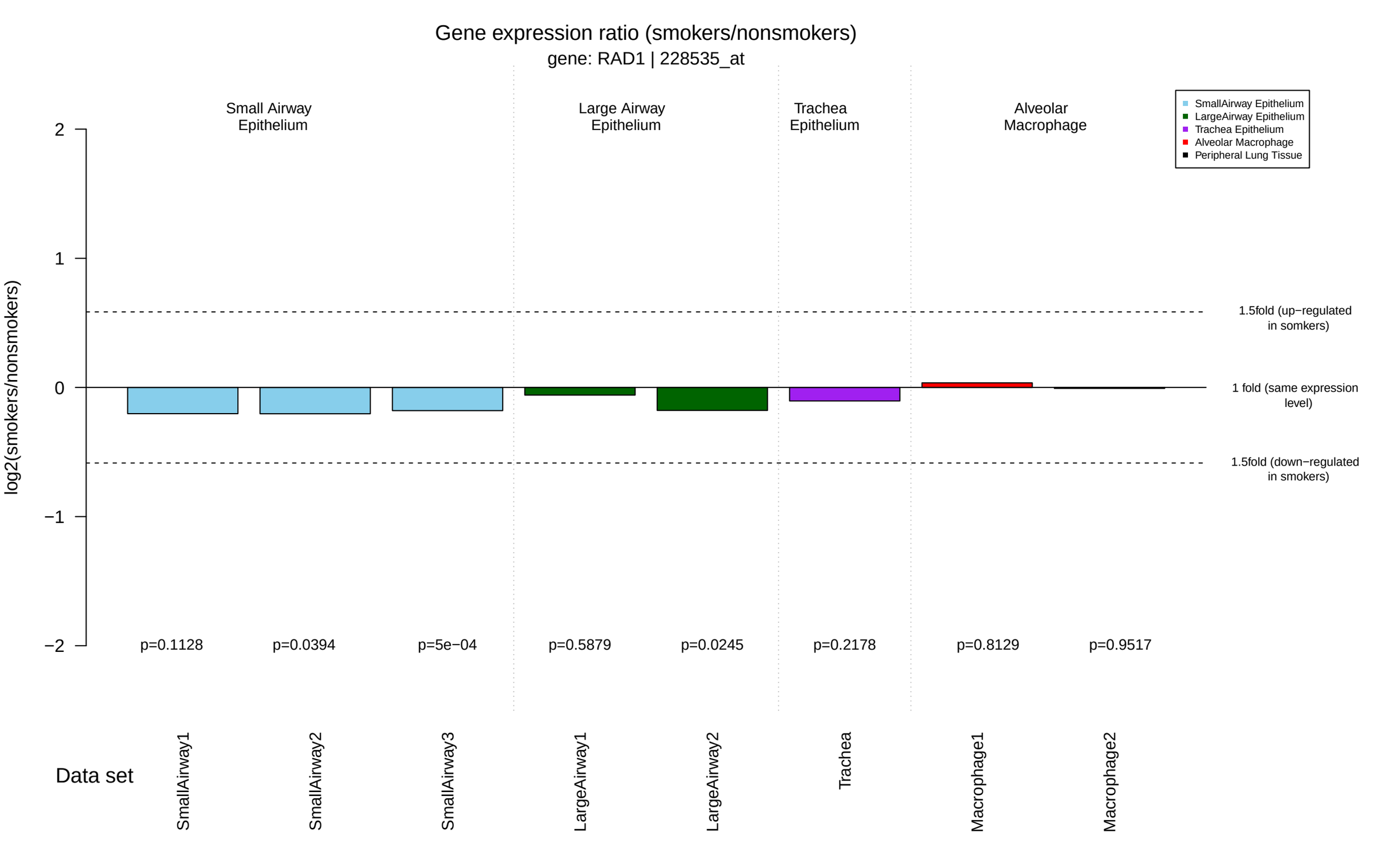
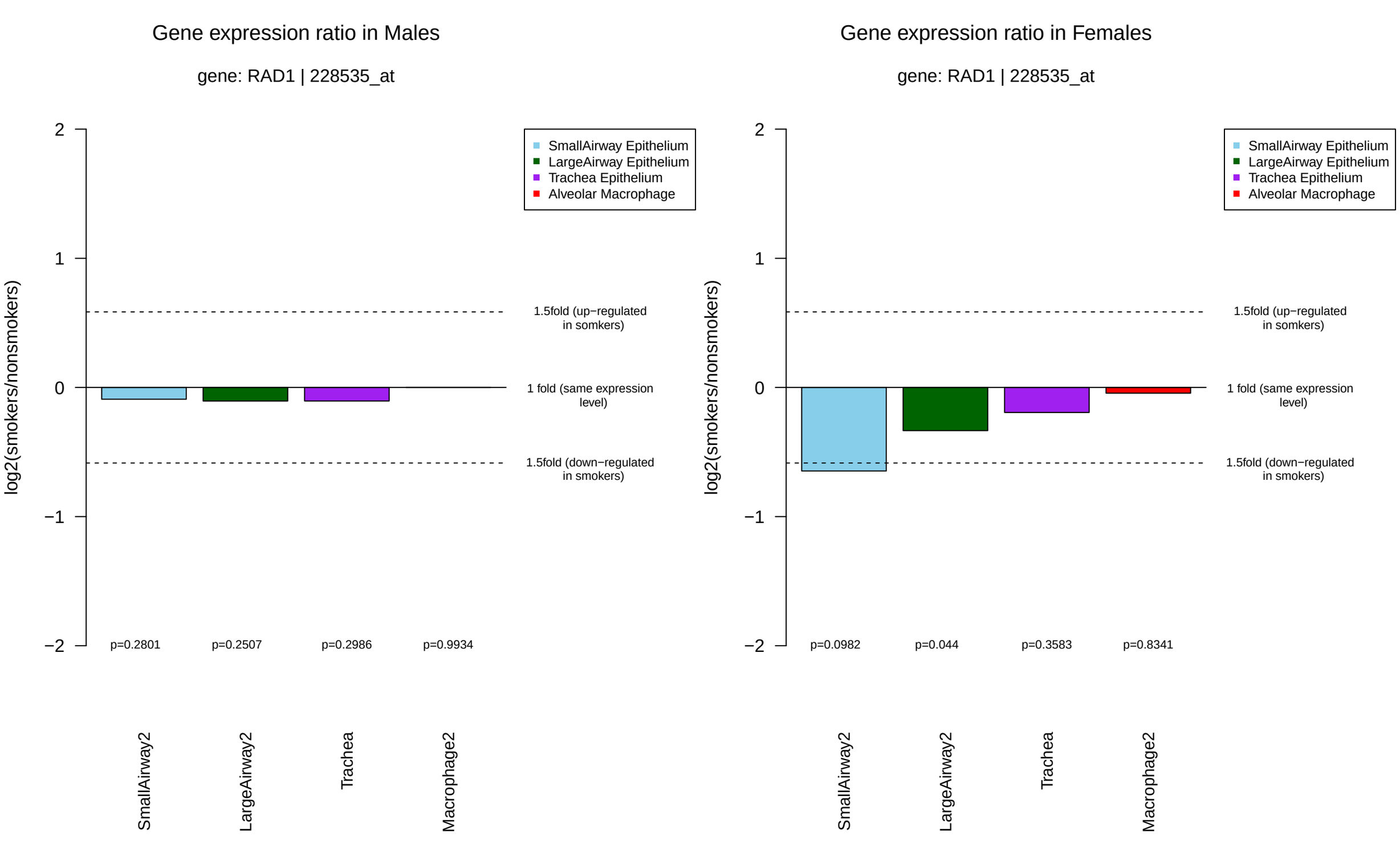
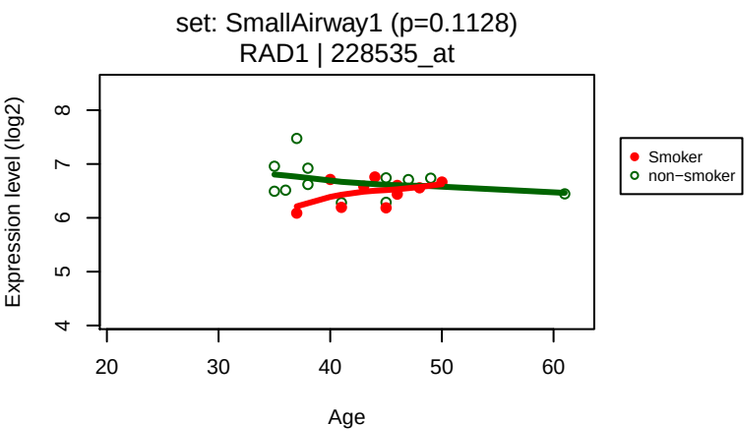
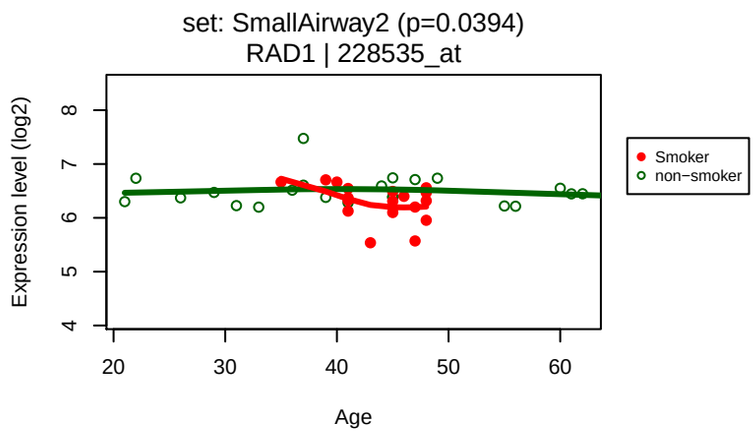
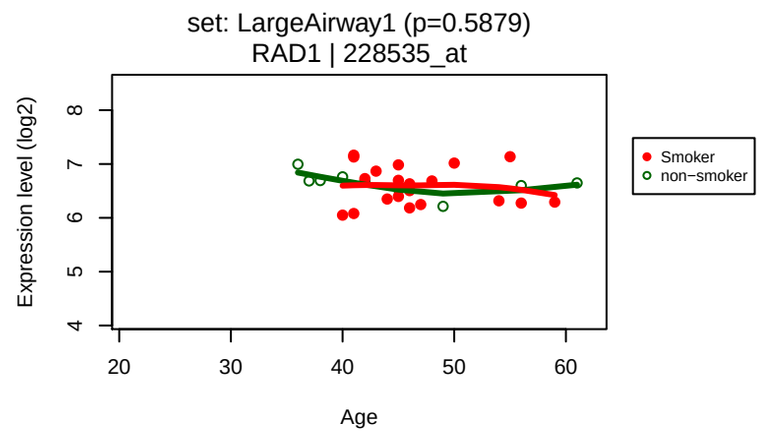

Gene expression of smokers and nonsmokers
gene: RAD1 | 228535_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1128
p=0.0394
p=5e−04
p=0.5879
p=0.0245
p=0.2178
p=0.8129
p=0.9517
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

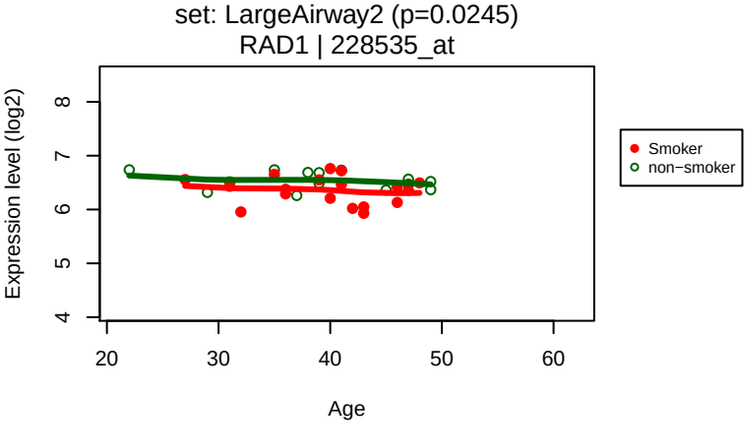

Gene expression ratio (smokers/nonsmokers)
gene: RAD1 | 228535_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1128
p=0.0394
p=5e−04
p=0.5879
p=0.0245
p=0.2178
p=0.8129
p=0.9517
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

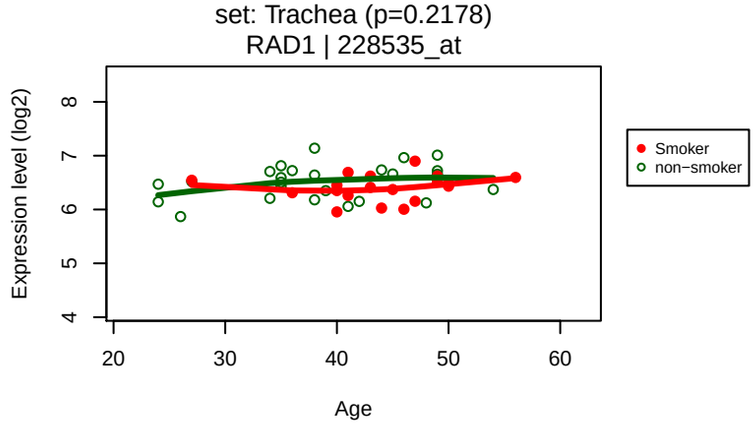

Gene expression ratio in Males
gene: RAD1 | 228535_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2801
p=0.2507
p=0.2986
p=0.9934
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAD1 | 228535_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0982
p=0.044
p=0.3583
p=0.8341
SmallAirway2
LargeAirway2
Trachea
Macrophage2

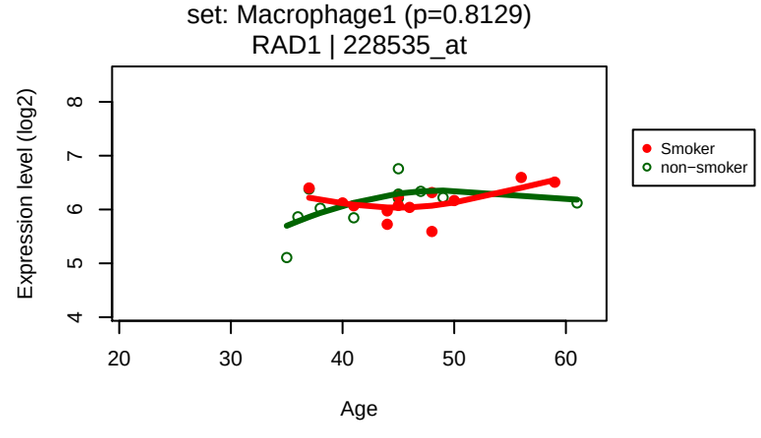

set: SmallAirway1 (p=0.1128)
RAD1 | 228535_at
Expression level (log2)
Age
Smoker
non−smoker

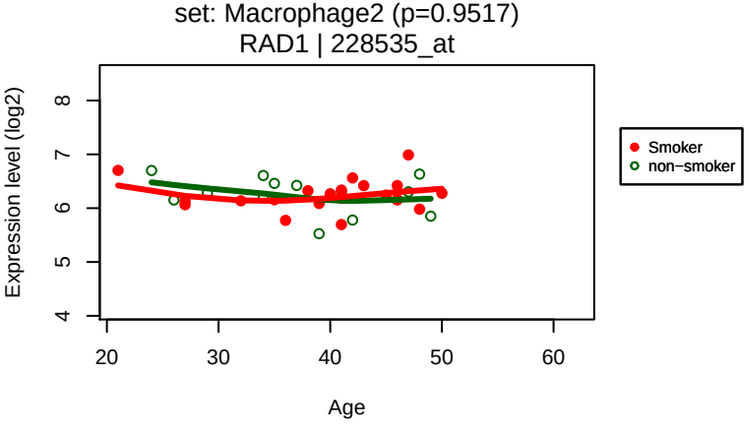

set: SmallAirway2 (p=0.0394)
RAD1 | 228535_at
Expression level (log2)
Age
Smoker
non−smoker

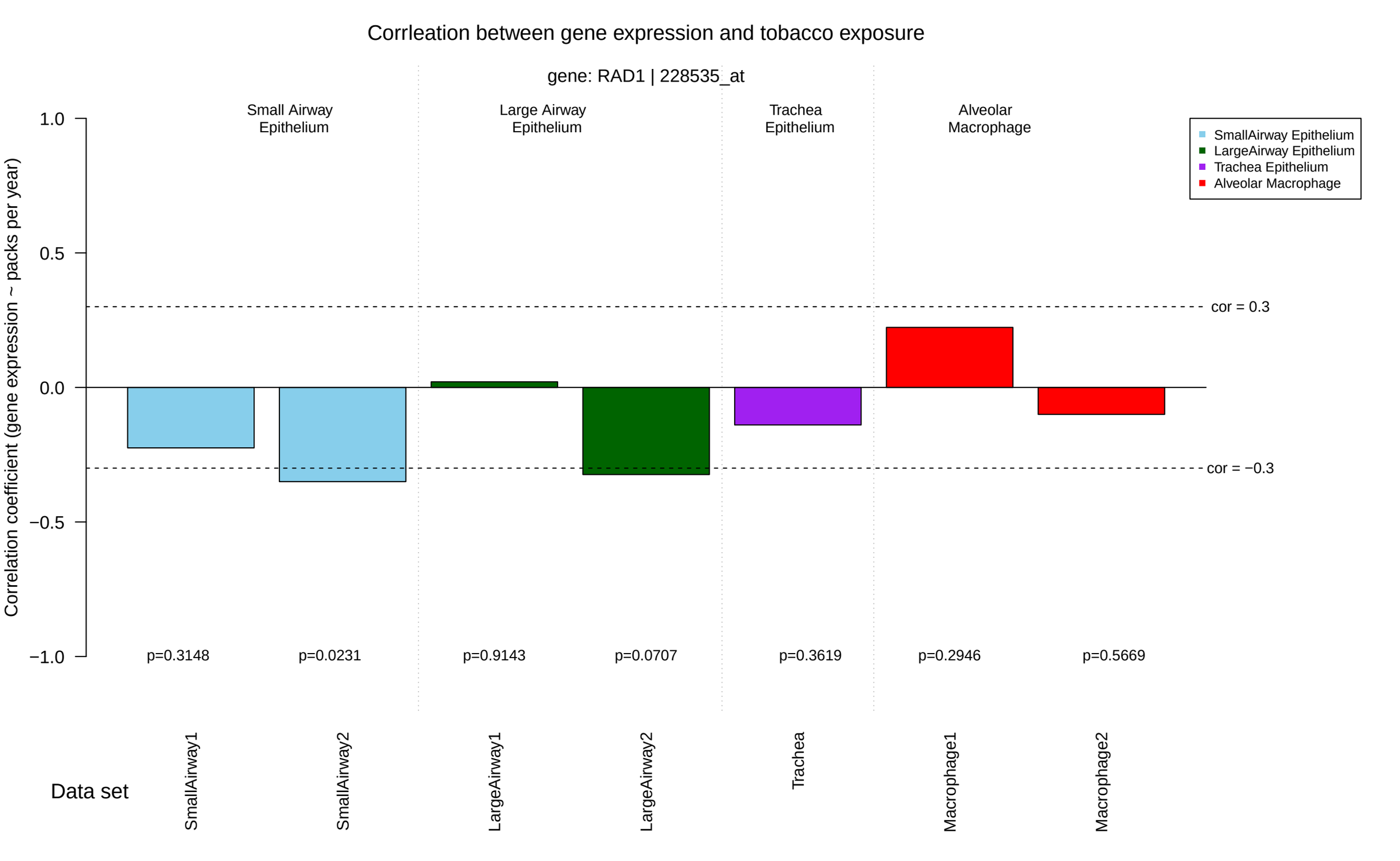

set: LargeAirway1 (p=0.5879)
RAD1 | 228535_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0245)
RAD1 | 228535_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2178)
RAD1 | 228535_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8129)
RAD1 | 228535_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9517)
RAD1 | 228535_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAD1 | 228535_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3148
p=0.0231
p=0.9143
p=0.0707
p=0.3619
p=0.2946
p=0.5669
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2